# From User Recognition to Activity Counting: An Identity-Agnostic Approach to Multi-User WiFi Sensing

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***Abstract*—WiFi Channel State Information (CSI) enables device-free human activity recognition, but multi-user methods typically assume a fixed set of known users and degrade on unseen individuals. We reformulate multi-user recognition as identity-agnostic activity counting, estimating how many users perform each activity without assigning actions to individuals. For a controlled comparison, identity-dependent predictions are converted to activity counts and evaluated with the same integer-valued metrics using a shared pipeline that differs only in output head and objective. On WiMANS, identity-agnostic counting achieves lower integer-count MAE than derived identity-dependent counts (0.0815 versus 0.0909 on a 0–5 count scale) despite using coarser supervision. This ordering holds across all eight environment–band conditions, while derived counts lead only in exact-match accuracy. Under leave-users-out evaluation, identity-dependent macro-F1 drops from 80.02% to 33.85%, its derived counts perform only slightly better than an all-zero predictor, and the exact-match advantage reverses, while identity-agnostic counting leads on all counting metrics. Silhouette analysis shows that identity-dependent embeddings cluster by user identity, whereas identity-agnostic embeddings cluster by activity composition. Activity counting therefore provides a more accurate and robust formulation for scene-level activity composition estimation without requiring a predefined user set.**

***Index Terms*—WiFi sensing, Channel State Information (CSI), Activity recognition, Multi-user, Activity counting, Generalization.**

## I. Introduction

Recent advances in wireless sensing have extended the capabilities of commodity WiFi infrastructures beyond traditional communication, transforming them into a pervasive and device-free modality for perceiving the physical world [1]–[3]. In particular, Channel State Information (CSI), which characterizes the fine-grained propagation properties of wireless signals across subcarriers and antennas, provides rich information about the interaction between radio signals and human motion. Compared with traditional vision-based or wearable sensing approaches, CSI-based systems offer several advantages such as non-intrusive sensing, reduced privacy concerns, and the ability to operate using widely deployed commodity infrastructures. These properties have motivated extensive research efforts toward leveraging CSI for a broad range of human-centric sensing tasks. Consequently, CSI has demonstrated promising performance in diverse applications such as device-free activity recognition [4], [5], gesture recognition [6], [7], fall detection [8], [9], respiration monitoring [10], and intrusion detection [11], [12]. Despite these significant advancements, many CSI-based sensing systems are primarily designed for single-user scenarios under controlled environments, while real-world deployments often involve multiple users interacting simultaneously within dynamic indoor environments.

Recognizing human activities from WiFi signals becomes considerably more difficult in multi-user scenarios, as concurrent body movements cause signal distortion, interference, and overlapping reflections in the wireless channel [13]. To address this complexity, existing studies adopt identity-dependent formulations that predict activities separately for each predefined subject [14]–[16]. These methods assume a fixed set of known users and frame recognition as a user-indexed prediction task. While effective under controlled conditions, such formulations limit practical deployment because real-world environments often include unseen users and a changing number of subjects. Current systems are typically evaluated using closed-set protocols where all subjects appear during training, raising uncertainty as to whether reported performance reflects genuine activity understanding or partially relies on user-specific signal characteristics. This uncertainty reflects a cost that is intrinsic to the formulation. Requiring a model to decide which user performs which activity forces it to encode user-specific signal characteristics, which are informative for enrolled individuals and misleading for anyone else. The requirement is also stronger than most deployments need, as these ask only what is happening in a scene rather than who is doing it.

In this work we remove that requirement. Instead of assigning activities to predefined individuals, we propose an identity-agnostic formulation that we refer to as activity counting: estimating the number of occurrences of each activity type at a given time, without associating actions with specific individuals. This shifts the task from user-indexed classification to scene-level activity composition estimation and removes the need for fixed user sets. We develop and evaluate both formulations on the WiMANS dataset [17], implemented on a common pipeline built around a standard convolutional backbone, so that any difference between them follows from the output structure and training objective rather than from architectural asymmetry. Because the two produce outputs of different kinds, user-wise predictions are converted into derived activity counts, which places them on identical metrics and makes the comparison possible at all. As the identity-dependent side of this pipeline substantially exceeds every baseline evaluated here, the comparison is between two strong systems rather than against a weakened control. The main contributions are as follows:

- We propose an identity-agnostic formulation for multi-user wireless activity analysis that replaces user-indexed prediction with scene-level activity counting, removing

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the need for a predefined set of users.
- Using matched count metrics, we show that identity-agnostic counting is more accurate and robust than counts derived from identity-dependent predictions across standard, unseen-user, and unseen-environment settings. Feature-space analyses reveal representations organized around activity composition rather than user identity, while linear probes on frozen features show that the counting advantage persists under an identical readout.
- We show that ImageNet-pretrained convolutional features transfer effectively to multi-user CSI projections, where four of the five CSI architectures fitted with the same counting head yield negative $R^2_{\text{int}}$. Permutation and from-scratch ablations suggest that performance depends primarily on pretrained filters and preserved temporal locality rather than precise projection geometry.

## II. Related Works

### A. Single-User Activity Recognition

Early CSI-based recognition systems relied on handcrafted signal features [4], [18], but deep learning approaches have largely replaced manual feature engineering. Architectures ranging from autoencoder-enhanced convolutional-recurrent networks [5], hybrid CNN-GRUs [19], attention-based sequence models [20], and transformers [21], [22] to pure 2D-CNN designs [23] have achieved high performance by capturing complex spatio-temporal patterns in the wireless channel. However, they are fundamentally formulated for single-user settings, whereas real-world environments often involve multiple people performing different activities simultaneously.

### B. Multi-User Activity Recognition

A common strategy in early multi-user WiFi sensing was to first separate user-specific signal components and then apply per-user activity classifiers. WISDOM [14], MultiTrack [24], and IMar [25] follow this general strategy using, respectively, signal sorting, multi-link profile reconstruction, and tensor decomposition to isolate user-specific information, while WiMU [26] instead generates virtual multi-user samples from single-user training data. Collectively, these systems demonstrate that multi-user recognition is feasible for several simultaneous users, but they are all evaluated on private datasets, which limits reproducibility and makes systematic comparison difficult.

More recent work has shifted toward publicly available benchmarks, with WiMANS [17] emerging as a common evaluation platform. On this dataset, MultiSenseX [15] studies joint localization and activity recognition in multi-occupant settings using a multi-label transformer design, Wang et al. [16] combine inception modules with attention mechanisms to capture both short-term and long-term patterns in CSI, and WiMAR [27] emphasizes the separation of dynamic and static signal components and processes the motion-related component using a CNN–GRU–attention pipeline. More recently, TSDBN [28] proposes a lightweight dual-branch design that processes a temporally compressed CSI representation through a convolution-augmented attention branch and a parallel 2D convolutional branch, and reports identity, activity, and location recognition on WiMANS under both bands. These studies provide a more comparable basis for evaluation, but they also share an important limitation, as performance is reported primarily in terms of accuracy. This is problematic for WiMANS because the dataset exhibits a strong class imbalance induced by the *null* label (represented as *nan* in the dataset codebase), which denotes the absence of a user in a given slot. Since multiple user slots may be unoccupied in the same sample, a model with weak discriminative ability can still achieve misleadingly high accuracy by over-predicting *null*. This makes accuracy an unreliable indicator of true recognition performance, and we argue that macro-F1 score provides a more faithful evaluation.

More recently, UN-2DCNN [29] reports state-of-the-art results, but because some preprocessing details are not fully specified, we were unable to reproduce the pipeline reliably and therefore exclude it from direct comparison.

Despite their methodological diversity, all of the above approaches share a common assumption: activities are predicted separately for each predefined individual, tying model outputs directly to a fixed set of known users. Although IMar includes action counting, this counting remains user-indexed and still depends on prior knowledge of user count together with successful per-user signal decomposition. A recent exception is AMAR [30], which formulates multi-user HAR as set prediction: the predictions of a set of query slots are matched to ground-truth activities by the Hungarian algorithm, so the output carries no identity indexing and activity counts can be obtained by aggregating the slots. Its slot count must nonetheless be fixed in advance, and its evaluation is confined to environment-specific standard splits, so the effect of removing identity supervision on cross-user generalization remains an open question. In contrast, this work reformulates the task as scene-level activity composition estimation, removing the dependence on user identity and count altogether. This enables more flexible generalization to settings in which the number and identity of active users may be unknown at inference time.

Beyond these specific studies, reproducibility remains a significant concern across the multi-user sensing literature. As highlighted by Guarino et al. [31], a large portion of published works lack essential experimental details, publicly available datasets, or accessible preprocessing code, making systematic comparison across studies challenging.

### C. Generalization Across Users and Environments

Generalization across unseen users and environments remains one of the central challenges in WiFi sensing. As identified by Wang et al. [32], three major factors hinder generalization: device heterogeneity, human body diversity, and environmental diversity. Even in single-user scenarios, differences in body shape, motion style, and room layout can produce substantial variation in the measured signal.

Existing work has mainly studied generalization along two axes: unseen users and unseen environments. For cross-user robustness, Widar3.0 [33] derives velocity profiles that capture kinematic characteristics independent of the performer or location. Meta-learning approaches such as MetaFormer [34] and Wi-Learner [35] further improve adaptation to unseen users under limited supervision, while CrossID [36] explicitly models

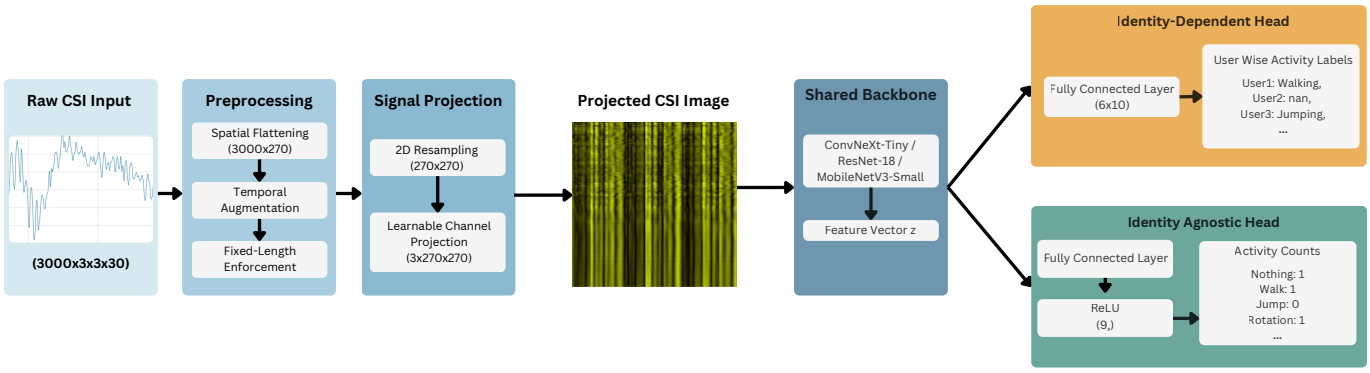


Fig. 1: Overview of the proposed multi-user CSI sensing framework. Raw CSI signals are preprocessed and projected into a 2D representation, processed by a shared pretrained backbone, and fed into either identity-dependent or identity-agnostic prediction heads.

identity-related information as a transferable representation. FewSense [37] pre-trains a feature extractor offline and classifies novel classes in unseen domains by cosine similarity, fine-tuning on a handful of labelled target samples, while WiTeacher [38] removes the need for target labels altogether by combining a Mean Teacher framework with generated target-like samples and adaptive label smoothing.

On the environment side, AirFi [39] proposes a domain generalization framework that extracts and augments common features across multiple training environments to enable recognition in unseen environments without target data. WiHARAN [40] aligns the joint distribution of features and labels across environments through adversarial learning, while DA-HAR [41] extracts environment-independent features using a dual adversarial network with pseudo-label prediction. Additionally, Zhang et al. [42] apply the Johnson-Lindenstrauss transform to generate differentially private representations for cross-environment sharing. What these methods have in common is that they counter domain shift with an additional resource: labelled or unlabelled samples from the target domain, data from several training environments, or hand-designed domain-invariant features that require particular hardware configurations. They are also all developed for single-user settings. The present work instead asks whether the choice of problem formulation itself affects sensitivity to unseen users, without any of these and in the multi-user setting.

## III. Dataset Description

All experiments were conducted on the publicly available WiMANS dataset [17], which is the first benchmark specifically designed for WiFi-based multi-user activity sensing. Unlike prior CSI datasets that contain only a single subject per sample, WiMANS captures simultaneous activities of multiple users within the same time window. It contains 11,286 CSI samples collected in three indoor environments: *classroom*, *meeting room*, and *empty room*. Each scene was recorded on both the 2.4 GHz and 5 GHz bands, so a single band provides 5,643 samples. Each sample has a duration of 3 seconds and includes between 0 and 5 simultaneously active users, drawn from a pool of six participants (anonymized as User 1–User 6). CSI is recorded using 3 transmit and 3 receive antennas over 30 subcarriers at a nominal rate of 1000 packets per second. Recordings vary slightly in length because of packet loss and are reflection-padded to 3000 packets where necessary, yielding an input dimension of $3000 \times 3 \times 3 \times 30$.

The dataset defines nine daily activities: *Nothing*, *Walking*, *Rotation*, *Jumping*, *Waving*, *Lying Down*, *Picking Up*, *Sitting Down*, and *Standing Up*. These range from sustained whole-body motion to brief localised gestures and to *Nothing*, which denotes a user present in the scene but stationary. Each sample is annotated with environment label, number of users, user identities, locations, and per-user activity labels. Activity labels are provided for six identity slots, with absent users marked as `nan` in the dataset codebase. We encode these as a tenth label, $\emptyset$, distinct from the *Nothing* activity above. The samples are distributed equally across the three environments.

Three properties of WiMANS are important for this work. First, multiple users may perform different activities simultaneously, creating overlapping signal reflections that make identity-dependent recognition challenging. Second, occupancy varies across samples and is balanced by design: the dataset contains equal numbers of samples for two, three, four, and five active users (15.79% each), twice as many single-user samples (31.58%), and a smaller set of empty scenes (5.26%), giving a mean of 2.53 active users. As these users are distributed over nine activity classes, most entries of the ground-truth count vector are zero. Third, the six-participant pool allows the users seen during training to be disjoint from those seen at test time, which is what the leave-users-out protocol of Section IV-D exercises.

## IV. Methodology

### A. Problem Formulation

Let $X \in \mathbb{R}^{T\times3\times3\times30}$ denote a CSI sample with $T = 3000$ temporal packets. Two formulations are considered: Identity-Dependent Activity Recognition (ID) and Identity-Agnostic Activity Counting (IA). In the identity-dependent activity recognition formulation, the task is defined as predicting the activity of each predefined user slot separately. Let $U = 6$ denote the number of user slots defined by the dataset, of which at most five are occupied in any scene. For each user $u \in \{1, \ldots, U\}$, the activity label is defined as $y_u \in \mathcal{A}_{\text{dep}} = \{\emptyset, a_1, a_2, \ldots, a_9\}$.

Writing $\mathbf{y}_u \in \{0,1\}^{10}$ for the one-hot encoding of $y_u$, the target for a given CSI sample is the matrix $Y \in \{0,1\}^{U\times 10}$ whose $u$-th row is $\mathbf{y}_u$. The model learns a mapping $f_\theta : X \to \hat{Y}$, where $\hat{Y} \in \mathbb{R}^{U\times 10}$ contains class scores for each user slot; the predicted label for slot $u$ is $\arg\max_j \hat{Y}_{u,j}$. The output is tied to a fixed user roster and therefore assumes consistent identity indexing in closed-set operation; the leave-users-out protocol deliberately violates this assumption.

In the identity-agnostic formulation, the task is to estimate the number of occurrences of each activity type within the scene, without associating actions with specific users. For a given CSI sample, the target output is $\mathbf{c} = [c_1, \ldots, c_9] \in \mathbb{N}_0^9$, where each element $c_k$ represents the number of users performing activity $a_k$ in that sample. The ground-truth count for class $a_k$ is computed as $c_k = \sum_{u=1}^{U} \mathbb{I}[y_u = a_k]$. For example, two users walking and one sitting down gives $c_{\text{walk}} = 2$, $c_{\text{sit_down}} = 1$, and zero elsewhere. The model learns a mapping $g_\theta : X \to \hat{\mathbf{c}}$, where $\hat{\mathbf{c}} \in \mathbb{R}^9_{\geq 0}$ are continuous non-negative predictions obtained through a regression head. For discrete counting evaluation, predictions are rounded element-wise to the nearest integer.

### B. CSI Representation, Projection, and Backbone

CSI amplitude measurements were used as the input representation. As the first preprocessing step, the spatial and frequency dimensions ($3 \times 3$ antenna pairs and 30 subcarriers) were flattened to obtain a time-feature matrix $X_f \in \mathbb{R}^{T\times 270}$, preserving temporal ordering. In this representation, the rows preserve the temporal packet sequence, while the columns encode the antenna-pair and subcarrier responses. Two-dimensional convolutional filters can therefore capture local temporal variations while aggregating information across the wireless channels. During training, temporal warping was applied with probability 0.5 by resampling the temporal axis using a scaling factor drawn from $[0.95, 1.05]$. All samples were then resized to a fixed length of 3000 packets by truncation or reflection padding. To transfer ImageNet-pretrained vision backbones to CSI sensing, the resulting matrix was interpreted as a single-channel 2D signal and resized with bicubic interpolation to $X_r \in \mathbb{R}^{270\times 270}$. This resolution preserves the flattened feature dimension without additional compression. A learnable $1 \times 1$ convolution then projected $X_r$ to $X_{rgb} \in \mathbb{R}^{3\times 270\times 270}$, producing a three-channel representation compatible with the pretrained convolutional networks. This projection does not assume semantic equivalence between CSI matrices and natural images; rather, it provides a regular two-dimensional structure that can be adapted by pretrained convolutional filters during end-to-end fine-tuning. Section V-G examines which of its two axes the resulting model actually relies on.

The projected tensor $X_{rgb}$ is processed by an ImageNet-1K-pretrained convolutional backbone. Three architectures with different capacity levels were evaluated: ConvNeXt-Tiny [43], ResNet-18 [44], and MobileNetV3-Small [45]. For each model, the classification layer was removed while retaining the convolutional feature extractor and global average pooling, yielding a feature vector $\mathbf{z} \in \mathbb{R}^d$. This shared representation was then passed to task-specific prediction heads. All backbones were fine-tuned end-to-end. The overall architecture is illustrated in Fig. 1.

### C. Task-Specific Prediction Heads and Objectives

Given the shared feature representation $\mathbf{z} \in \mathbb{R}^d$ extracted by the backbone, the final prediction layer and training objective differ depending on the task formulation. The source code will be released upon acceptance.

**Identity-dependent recognition:** The objective was to predict the activity label for each of the $U = 6$ predefined user slots across 10 classes (including the $\emptyset$ class). A fully connected layer mapped the feature vector to class logits: $\mathbf{z} \to \hat{Y} \in \mathbb{R}^{U\times 10}$. To mitigate the class imbalance caused by the frequent $\emptyset$ class, the model was optimized using a focal loss [46] ($\gamma = 2.0$) applied independently to each user slot. Under the leave-users-out protocol the head has three slots rather than six.

**Identity-agnostic counting:** A single fully connected layer mapped the features to continuous count predictions for the 9 distinct activities (excluding $\emptyset$): $\mathbf{z} \to \hat{\mathbf{c}} \in \mathbb{R}^9$. A ReLU activation ensured non-negative outputs, and the model was trained with a Mean Squared Error (MSE) objective between predicted and ground-truth activity counts.

### D. Implementation and Evaluation Details

The models were trained on an NVIDIA RTX A4000 GPU for 50 epochs using AdamW (weight decay $10^{-2}$), a batch size of 16, and gradient clipping (max norm 1.0). A two-stage learning rate schedule was applied: a linear warm-up during the first 10% of epochs followed by cosine annealing. The initial learning rate was set to $10^{-3}$ for the projection module and $10^{-4}$ for the backbone and prediction heads. Two baselines used paper-specific training schedules rather than the common 50-epoch setup: TSDBN for 200 epochs under the same optimiser, and AMAR for 300 epochs with Adam following its authors' configuration. The remaining baselines follow the configuration above.

**Evaluation Metrics:** Identity-dependent recognition was evaluated with macro-F1, which is preferred to overall accuracy because of the class imbalance induced by the frequent $\emptyset$ label. For scene-level comparison, the user-wise predictions were converted into derived counts. Let $\hat{y}_u = \arg\max_j \hat{Y}_{u,j}$ denote the predicted class for user slot $u$; the derived count for activity $a_k$ was computed as $\hat{c}_k^{\text{dep}} = \sum_{u=1}^{U} \mathbb{I}[\hat{y}_u = a_k]$. Slots predicted as $\emptyset$ therefore contributed to none of the nine counts, whereas *Nothing* is one of them. The same conversion was applied to all identity-dependent models, including the re-implemented baselines.

Identity-agnostic counting produces continuous predictions, which were evaluated using $\text{MAE}_{\text{cont}}$ and $R^2_{\text{cont}}$. For direct comparison with the integer-valued derived counts, these predictions were rounded to the nearest integer; metrics computed from integer-valued predictions are denoted $\text{MAE}_{\text{int}}$ and $R^2_{\text{int}}$. MAE was computed over all activity–sample entries of the count matrix, whereas $R^2$ was computed separately for each of the nine activity dimensions and averaged uniformly across them. Cell-wise accuracy measures the proportion of individual activity counts predicted exactly, whereas exact-match accuracy requires the complete activity-count vector

TABLE I

PERFORMANCE AND COMPLEXITY UNDER THE STANDARD PROTOCOL (5 GHZ, ALL ENVIRONMENTS). COMPLEXITY MEASUREMENTS CORRESPOND TO THE IDENTITY-AGNOSTIC CONFIGURATIONS. BOLD DENOTES THE BEST VALUE PER COLUMN. CPU LATENCY IS MEASURED PER SAMPLE USING FP32, BATCH SIZE 1, AND A SINGLE THREAD.

| | ID Recognition | ID-Derived Counts | | | | IA Counting | | | | Complexity | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Macro-F1 (%) | Cell Acc. (%) | Exact Match (%) | $\mathrm{MAE}_{\mathrm{int}}$ | $R^2_{\mathrm{int}}$ | Cell Acc. (%) | Exact Match (%) | $\mathrm{MAE}_{\mathrm{int}}$ | $R^2_{\mathrm{int}}$ | Params (M) | MACs (G) | Latency (ms) |
| 2D-CNN [23] | 26.47 ± 0.68 | 70.54 ± 0.07 | 11.43 ± 0.78 | 0.3693 ± 0.0049 | −0.3895 ± 0.0175 | 74.03 ± 0.38 | 6.11 ± 0.62 | 0.3028 ± 0.0066 | −0.0316 ± 0.0412 | 0.8874 | 0.8421 | 15.1 |
| CNN-LSTM [47] | 25.83 ± 0.75 | 69.51 ± 0.09 | 10.83 ± 1.03 | 0.3876 ± 0.0057 | −0.4838 ± 0.0277 | 76.19 ± 0.16 | 5.46 ± 1.17 | 0.2923 ± 0.0199 | −0.0991 ± 0.1563 | 5.3682 | 0.8587 | 32.1 |
| ABLSTM [20] | 27.68 ± 0.78 | 71.17 ± 0.51 | 11.54 ± 1.34 | 0.3588 ± 0.0082 | −0.3327 ± 0.0196 | 77.22 ± 0.52 | 5.67 ± 1.08 | 0.2780 ± 0.0088 | −0.0272 ± 0.1010 | 4.2706 | 0.3936 | 157.4 |
| THAT [21] | 41.28 ± 1.48 | 77.85 ± 0.22 | 24.95 ± 0.78 | 0.2691 ± 0.0052 | 0.0447 ± 0.0082 | 78.35 ± 0.42 | 4.99 ± 0.50 | 0.2849 ± 0.0002 | −0.2002 ± 0.0099 | 4.8885 | 0.5492 | 24.6 |
| TSDBN [28] | 37.38 ± 1.38 | 74.78 ± 0.52 | 23.98 ± 1.88 | 0.3114 ± 0.0095 | −0.1637 ± 0.0531 | 77.16 ± 0.57 | 17.18 ± 1.25 | 0.2643 ± 0.0027 | 0.1219 ± 0.0250 | 1.3612 | 0.1125 | **7.1** |
| AMAR w/o RVQ [30] | – | – | – | – | – | 83.76 ± 0.34 | 44.73 ± 1.68 | 0.1824 ± 0.0030 | 0.4244 ± 0.0286 | **0.3117** | 0.1947 | 14.6 |
| Ours (MobileNetV3-Small) | 65.97 ± 0.55 | 85.80 ± 0.34 | 47.80 ± 1.33 | 0.1601 ± 0.0036 | 0.5008 ± 0.0221 | 86.69 ± 0.22 | 37.11 ± 0.76 | 0.1425 ± 0.0035 | 0.5947 ± 0.0100 | 0.9322 | **0.0840** | 9.4 |
| Ours (ResNet-18) | 66.14 ± 0.21 | 86.21 ± 0.14 | 51.37 ± 0.85 | 0.1552 ± 0.0025 | 0.5174 ± 0.0230 | 87.05 ± 0.83 | 37.23 ± 4.38 | 0.1436 ± 0.0113 | 0.5514 ± 0.0632 | 11.1811 | 2.7454 | 57.8 |
| Ours (ConvNeXt-Tiny) | **80.02 ± 0.31** | **91.71 ± 0.19** | **68.50 ± 0.49** | **0.0909 ± 0.0028** | **0.7308 ± 0.0058** | **92.25 ± 0.11** | **57.75 ± 0.85** | **0.0815 ± 0.0018** | **0.7769 ± 0.0111** | 27.8255 | 5.9958 | 117.0 |

of a sample to be correct; both were computed from integer-valued predictions. All conversions were applied only during evaluation and did not alter either training objective.

**Data Splits and Baselines:** Generalization was evaluated across three protocols: (1) *Standard Split* (training and testing on the same environments/users), (2) *Leave-Users-Out (LUO)*, where models were trained on three users and evaluated on three disjoint users, and (3) *Leave-One-Environment-Out (LOEO)*. Standard and LOEO retain the six global identity slots. LUO instead uses a three-slot head: the disjoint training and test identities are independently mapped, in list order, to local slots 1–3. Every output unit therefore receives training supervision, including $\emptyset$ labels; only its associated physical identity changes at test time, avoiding unused output units as a confound. The Standard and LOEO partitions have mean test occupancies of 2.56 and 2.53, whereas the LUO partitions have 1.25, so counting errors are compared between formulations within a protocol rather than across protocols. A predictor emitting zero for every activity attains an $\mathrm{MAE}_{\mathrm{int}}$ of 0.2849, 0.1389, and 0.2807 on the Standard, LUO, and LOEO partitions respectively. All experiments were repeated over three data partitions and are reported as mean ± SD.

The evaluated formulations were benchmarked against five representative CSI sensing architectures: 2D-CNN [23], CNN-LSTM [47], ABLSTM [20], THAT [21], and TSDBN [28], together with AMAR without RVQ [30], the stronger of the two configurations reported by its authors, as an additional counting-oriented baseline. All models received the same input matrix $X_f$ and the same Standard-protocol partitions, and each then applies the input transformation specified in its original work. The domain-shift protocols were applied only to the two formulations within the proposed pipeline. Except for AMAR, all were fitted with the prediction heads and objectives described above. AMAR retains its own set-prediction objective and augmentations, with activity counts obtained by aggregating its six query slots at evaluation time.

Computational complexity was evaluated in terms of total parameter count and multiply–accumulate operations (MACs) for a single forward pass, together with mean inference latency per sample. Latency was measured on an Intel Xeon W-2223 CPU (3.6 GHz) in FP32 at batch size 1 with a single thread, as an average over 500 consecutive forward passes following 50 warm-up passes. To examine the learned representations, 768-dimensional feature vectors were extracted from the final pooled layer preceding the prediction heads, using the single-user test samples of the Standard partitions, since attributing a sample to a user identity is only well defined when one user is active. Silhouette scores were computed in this space under two clustering criteria, grouping samples either by the identity of the active user or by their activity-count vector, which for single-user samples reduces to the activity performed, and the same vectors were projected to two dimensions with t-SNE for visualisation. Identical samples were used for both formulations.

To separate the contribution of the representation from that of the output head, a linear count probe was fitted on frozen features. For each formulation the projection module and backbone were frozen after training and a single linear layer with ReLU was trained for 50 epochs on the same MSE counting objective and partitions, without augmentation. The two probes therefore share an identical readout and differ only in the representation they receive.

Uncertainty in the cross-formulation comparison was assessed with a paired sample-level bootstrap over 10,000 resamples, clustered by sample identifier, on the per-sample difference between the derived and identity-agnostic count MAE. Unlike the across-partition SD, this quantifies test-sample rather than training-split variability.

## V. EXPERIMENTAL RESULTS

### A. Backbone Selection and Comparison with Existing Models

Three ImageNet-pretrained backbones were evaluated under both formulations (Table I). ConvNeXt-Tiny achieved the strongest predictive performance, reaching a macro-F1 of 80.02 and an $\mathrm{MAE}_{\mathrm{int}}$ of 0.0815, while no other backbone reached 67 in macro-F1 or fell below 0.1425 in $\mathrm{MAE}_{\mathrm{int}}$. It was therefore used for all subsequent experiments.

The five identity-dependent baselines reached macro-F1 scores between 25.83 and 41.28; the proposed model exceeded the highest of these, THAT, by 38.7 points. Their derived counts yielded $R^2_{\mathrm{int}} \leq 0.045$, four of them negative. Under the identity-agnostic formulation, AMAR w/o RVQ produced the strongest baseline count estimates ($\mathrm{MAE}_{\mathrm{int}} = 0.1824$, $R^2_{\mathrm{int}} = 0.4244$); the proposed model reduced this error by 55%.

TSDBN followed at 0.2643 with a positive but small $R^2_{\mathrm{int}}$ of 0.1219, while the remaining four architectures reached $\mathrm{MAE}_{\mathrm{int}} \geq 0.2780$ with negative $R^2_{\mathrm{int}}$.

TABLE II

INTEGER-COUNT RESULTS ACROSS PROTOCOLS, ENVIRONMENTS, AND BANDS (MEAN ± SD OVER THREE DATA PARTITIONS). DOMAIN-SHIFT EXPERIMENTS USE 5 GHZ AND ALL ENVIRONMENTS. COMPARISONS ARE WITHIN EACH ROW; BOLD DENOTES THE BETTER COUNTING RESULT. MACRO-F1 APPLIES ONLY TO ID RECOGNITION.

| | | ID Recognition | ID-Derived Counts | | | | IA Counting | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Setting | Band | Macro-F1 (%) | Cell Acc. (%) | Exact Match (%) | $\mathrm{MAE}_{\mathrm{int}}$ | $R^2_{\mathrm{int}}$ | Cell Acc. (%) | Exact Match (%) | $\mathrm{MAE}_{\mathrm{int}}$ | $R^2_{\mathrm{int}}$ |
| *Standard protocol* | | | | | | | | | | |
| All environments | 5 GHz | 80.02 ± 0.31 | 91.71 ± 0.19 | **68.50 ± 0.49** | 0.0909 ± 0.0028 | 0.7308 ± 0.0058 | **92.25 ± 0.11** | 57.75 ± 0.85 | **0.0815 ± 0.0018** | **0.7769 ± 0.0111** |
| | 2.4 GHz | 49.02 ± 0.35 | 79.83 ± 0.27 | **30.97 ± 1.68** | 0.2313 ± 0.0005 | 0.2391 ± 0.0321 | **82.25 ± 0.42** | 24.68 ± 1.29 | **0.1934 ± 0.0055** | **0.4219 ± 0.0101** |
| Classroom | 5 GHz | 71.52 ± 1.56 | 88.68 ± 0.86 | **59.68 ± 2.43** | 0.1245 ± 0.0089 | 0.6053 ± 0.0378 | **90.26 ± 1.15** | 50.66 ± 3.68 | **0.1028 ± 0.0129** | **0.7012 ± 0.0177** |
| | 2.4 GHz | 45.10 ± 1.01 | 79.04 ± 0.25 | **25.82 ± 1.77** | 0.2409 ± 0.0010 | 0.1373 ± 0.0130 | **82.29 ± 0.33** | 21.31 ± 2.40 | **0.1914 ± 0.0033** | **0.3890 ± 0.0369** |
| Empty room | 5 GHz | 73.12 ± 1.49 | 89.38 ± 0.87 | **59.59 ± 4.23** | 0.1164 ± 0.0096 | 0.6297 ± 0.0267 | **91.35 ± 0.73** | 53.58 ± 4.12 | **0.0902 ± 0.0094** | **0.7437 ± 0.0274** |
| | 2.4 GHz | 41.17 ± 2.38 | 77.52 ± 0.39 | **25.11 ± 1.26** | 0.2669 ± 0.0027 | −0.0121 ± 0.0653 | **80.03 ± 0.40** | 16.98 ± 0.96 | **0.2221 ± 0.0035** | **0.2601 ± 0.0489** |
| Meeting room | 5 GHz | 77.22 ± 0.88 | 89.92 ± 0.35 | **62.16 ± 1.34** | 0.1102 ± 0.0026 | 0.6623 ± 0.0064 | **91.46 ± 0.28** | 54.11 ± 1.16 | **0.0899 ± 0.0022** | **0.7421 ± 0.0121** |
| | 2.4 GHz | 39.91 ± 0.91 | 75.77 ± 0.92 | **21.04 ± 1.01** | 0.2837 ± 0.0132 | −0.0120 ± 0.0546 | **79.29 ± 1.08** | 15.65 ± 0.92 | **0.2288 ± 0.0088** | **0.2650 ± 0.0026** |
| *Domain shift* | | | | | | | | | | |
| Leave-users-out | 5 GHz | 33.85 ± 5.13 | 87.48 ± 1.19 | 20.68 ± 2.76 | 0.1335 ± 0.0137 | −0.1694 ± 0.2078 | **93.04 ± 1.62** | **54.24 ± 8.88** | **0.0707 ± 0.0160** | **0.4314 ± 0.1519** |
| Leave-one-environment-out | 5 GHz | 38.60 ± 4.45 | 80.03 ± 0.91 | 14.78 ± 0.97 | 0.2469 ± 0.0075 | 0.0610 ± 0.0581 | **84.07 ± 0.84** | **24.24 ± 3.13** | **0.1844 ± 0.0052** | **0.3799 ± 0.0117** |

A predictor emitting zero for every activity attains $\mathrm{MAE}_{\mathrm{int}} = 0.2849$ under the Standard protocol, 0.1389 under LUO, and 0.2807 under LOEO.

Parameter counts and MACs spanned roughly two orders of magnitude across models, whereas latency varied by a factor of about 20, and the three measures did not rank the models consistently. AMAR w/o RVQ had the fewest parameters, MobileNetV3-Small the fewest MACs, and TSDBN the lowest latency at 7.1 ms. MobileNetV3-Small required 71× fewer MACs than ConvNeXt-Tiny but was only 12.4× faster (9.4 against 117.0 ms per sample). The clearest discrepancy was ABLSTM, the slowest model overall at 157.4 ms despite requiring 15× fewer MACs than ConvNeXt-Tiny, reflecting the sequential computation of its recurrent layers.

### B. Counting Accuracy under Matched Metrics

Under the Standard protocol the identity-dependent model reached a macro-F1 of $80.02 \pm 0.31$, which serves as the closed-set reference for Section V-D. On matched count metrics (Table II), identity-agnostic counting was more accurate on three of the four counting metrics: $\mathrm{MAE}_{\mathrm{int}}$ of $0.0815 \pm 0.0018$ against $0.0909 \pm 0.0028$, cell-wise accuracy of 92.25% against 91.71%, and $R^2_{\mathrm{int}}$ of 0.7769 against 0.7308. The derived counts led on exact-match accuracy, 68.50% against 57.75%, for reasons discussed in Section VI-C. Before rounding, the identity-agnostic predictions yielded $\mathrm{MAE}_{\mathrm{cont}} = 0.1060 \pm 0.0021$ and $R^2_{\mathrm{cont}} = 0.8134 \pm 0.0026$. The two formulations also differed in how their errors were distributed: among samples with at least one incorrect entry, the derived counts contained 2.4 incorrect entries on average against 1.7 for identity-agnostic counting. The paired bootstrap yielded $\Delta = 0.0095$ in favour of identity-agnostic counting (95% CI 0.0051–0.0139).

IA counting achieved lower MAE than ID-derived counting with all three backbones evaluated in our framework. TSDBN showed the same pattern and was the only identity-dependent baseline whose IA counts attained a positive $R^2_{\mathrm{int}}$. For the remaining identity-dependent baselines, both formulations performed close to or below the all-zero predictor, making their relative ordering less informative. AMAR does not enter this comparison, since its set-prediction output has no identity-dependent counterpart. These results indicate that the IA advantage is not specific to a particular backbone.

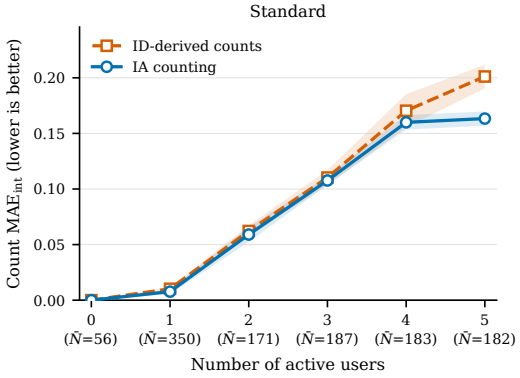


Fig. 2: Integer-count MAE by active-user count (5 GHz, all environments, Standard protocol; mean ± SD over three data partitions). $\bar{N}$ denotes the mean test-sample count per occupancy level.

### C. Effect of Environment, Frequency Band, and Scene Occupancy

Combined training outperformed environment-specific training under both formulations (Table II): macro-F1 reached 80.02 against 71.52 to 77.22, and $\mathrm{MAE}_{\mathrm{int}}$ 0.0815 against 0.0899 to 0.1028 for the identity-agnostic model. Performance was lower at 2.4 GHz in every environment, and the two formulations were affected to different degrees: $R^2_{\mathrm{int}}$ was negative for the derived counts in the empty and meeting rooms, whereas the identity-agnostic model retained 0.2601 and 0.2650. The ordering of Section V-B repeated in all eight environment–band conditions: identity-agnostic counting was more accurate on every count metric except exact match. Counting error increased monotonically with occupancy (Fig. 2); the two curves remained within 0.011 of each other up to four active users and separated at five, where the identity-agnostic model reached 0.163 against 0.201.

### D. Performance under User and Environment Shift

Because the protocol-specific test distributions differ, counting metrics are compared between formulations within each

TABLE III
SILHOUETTE SCORES IN THE 768-DIMENSIONAL EMBEDDING SPACE (5 GHZ, ALL ENVIRONMENTS, STANDARD PROTOCOL, SINGLE-USER SAMPLES; MEAN ± SD OVER THREE DATA PARTITIONS).

| Clustering criterion | ID Recognition | IA Counting |
|---|---|---|
| User identity | $0.345 \pm 0.013$ | $-0.037 \pm 0.004$ |
| Activity composition | $0.244 \pm 0.016$ | $0.669 \pm 0.019$ |

protocol (Table II).

Under LUO, the macro-F1 of the identity-dependent model decreased from 80.02 to $33.85 \pm 5.13$. On the same test samples, identity-agnostic counting was better on all four counting metrics, with an $\mathrm{MAE}_{\mathrm{int}}$ of 0.0707 against 0.1335, an exact-match accuracy of 54.24% against 20.68%, and an $R^2_{\mathrm{int}}$ of 0.4314 against $-0.1694$. The exact-match ordering was thus reversed with respect to the Standard protocol. Before rounding, the identity-agnostic predictions yielded $\mathrm{MAE}_{\mathrm{cont}} = 0.0899 \pm 0.0199$ and $R^2_{\mathrm{cont}} = 0.5199 \pm 0.1160$.

Under LOEO, where the identity-dependent macro-F1 was $38.60 \pm 4.45$, the identity-agnostic model was again better on all four metrics, with an $\mathrm{MAE}_{\mathrm{int}}$ of 0.1844 against 0.2469.

Normalizing MAE by each protocol's all-zero-predictor error provides a distribution-relative comparison across protocols. Under the Standard protocol, IA and ID-derived MAEs were 0.29 and 0.32 times the zero-predictor error, respectively. Both ratios increased under domain shift, to 0.51 and 0.96 under LUO and to 0.66 and 0.88 under LOEO. Thus, under user shift, the ID-derived MAE (0.1335) nearly equalled the zero-predictor MAE (0.1389), whereas IA counting produced approximately half that error. The paired bootstrap gave $\Delta = \mathrm{MAE}_{\mathrm{ID}} - \mathrm{MAE}_{\mathrm{IA}} = 0.0095$ (95% CI: 0.0051–0.0139) under Standard, increasing to 0.0629 (0.0595–0.0662) under LUO and 0.0625 (0.0584–0.0663) under LOEO, with the Standard interval disjoint from both.

### *E. Activity-Level Error Structure*

The confusion matrix of the identity-dependent model (Fig. 3) shows near-perfect recovery of ∅ (0.99) and diagonal values between 0.72 and 0.86 for the nine activities. Residual mass was concentrated in approximately symmetric pairs, principally *sit down–stand up* (0.10 in both directions) and *rotation–walk* (0.11 and 0.09); *Nothing* was additionally confused with *walk* and *stand up*.

Per-activity results provided a finer-grained view of the aggregate pattern (Fig. 4). Under the Standard protocol all nine activities remained below 0.12 for both formulations, the identity-agnostic model attaining lower error for five, most clearly *walk* at 0.0307 against 0.0650, and the derived counts attaining lower error for the remaining four by at most 0.006. Under LUO the identity-agnostic model attained lower error for all nine, with margins between 0.0384 and 0.0937, and the standard deviations of the derived counts increased markedly.

### *F. Feature-Space Characterization*

Cosine-distance silhouette scores (Table III) indicate that the identity-dependent embeddings were organized more strongly by user identity than by activity composition (0.345 versus 0.244), whereas the identity-agnostic embeddings showed the reverse ($-0.037$ versus 0.669). The negative identity-agnostic user score indicates that samples lay on average closer to other users' samples than to their own. The t-SNE projections qualitatively support this pattern (Fig. 5): users occupy largely separate regions under identity-dependent recognition but are substantially intermixed under identity-agnostic counting.

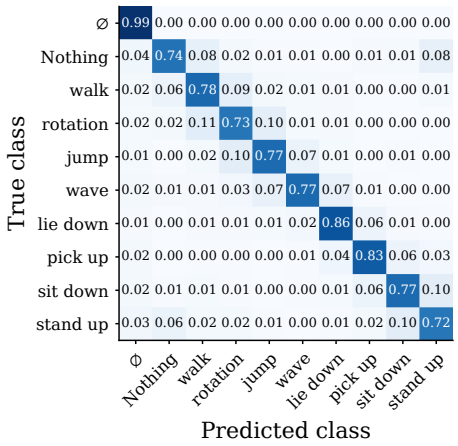


Fig. 3: Row-normalized identity-dependent confusion matrix averaged over three data partitions (5 GHz, all environments, Standard protocol).

TABLE IV
IDENTITY-AGNOSTIC ABLATION RESULTS (5 GHZ, ALL ENVIRONMENTS, STANDARD PROTOCOL; MEAN ± SD OVER THREE DATA PARTITIONS).

| Configuration | $\mathrm{MAE}_{\mathrm{int}}$ | Exact Match (%) | $R^2_{\mathrm{int}}$ |
|---|---|---|---|
| Proposed | $0.0815 \pm 0.0018$ | $57.75 \pm 0.85$ | $0.7769 \pm 0.0111$ |
| $224 \times 224$ resolution | $0.0826 \pm 0.0027$ | $58.34 \pm 1.29$ | $0.7734 \pm 0.0031$ |
| Bilinear interpolation | $0.0817 \pm 0.0017$ | $57.72 \pm 0.28$ | $0.7755 \pm 0.0071$ |
| Channel repetition | $0.0823 \pm 0.0029$ | $57.99 \pm 1.30$ | $0.7731 \pm 0.0054$ |
| Without temporal warping | $0.0900 \pm 0.0018$ | $55.62 \pm 1.17$ | $0.7488 \pm 0.0049$ |
| Feature-axis permutation | $0.0898 \pm 0.0024$ | $56.98 \pm 1.18$ | $0.7522 \pm 0.0045$ |
| Temporal-axis permutation | $0.2035 \pm 0.0020$ | $31.18 \pm 0.80$ | $0.3721 \pm 0.0084$ |
| From scratch | $0.2849 \pm 0.0002$ | $4.99 \pm 0.50$ | $-0.2002 \pm 0.0099$ |

The linear count probe separates the two contributions. Under the Standard protocol the probe on identity-agnostic features attained $\mathrm{MAE}_{\mathrm{int}} = 0.0830 \pm 0.0022$ and $R^2_{\mathrm{int}} = 0.7733 \pm 0.0084$, against $0.1121 \pm 0.0022$ and $0.6932 \pm 0.0195$ on identity-dependent features. The identity-agnostic probe recovered almost all of the end-to-end performance (0.0815), so the linear readout was not the limiting factor. Under LUO the gap persisted, $0.0865 \pm 0.0171$ against $0.1120 \pm 0.0146$, with the identity-dependent probe reaching a low and highly variable $R^2_{\mathrm{int}}$ of $0.0830 \pm 0.1611$.

### *G. Ablation Study*

Spatial resolution, interpolation method, and channel projection strategy left performance essentially unchanged, with $\mathrm{MAE}_{\mathrm{int}}$ differences of at most 0.0011, small relative to the observed across-partition variation (Table IV). Removing temporal warping increased $\mathrm{MAE}_{\mathrm{int}}$ from 0.0815 to 0.0900.

Two further variants permuted one axis of the resized CSI representation using the same fixed, one-to-one mapping for all training and test samples. This retained all values but rearranged their spatial neighborhoods. Permuting the antenna–subcarrier axis increased $\mathrm{MAE}_{\mathrm{int}}$ to 0.0898, a change comparable to removing temporal warping, whereas permuting the temporal axis increased it to 0.2035 and reduced exact-match accuracy to

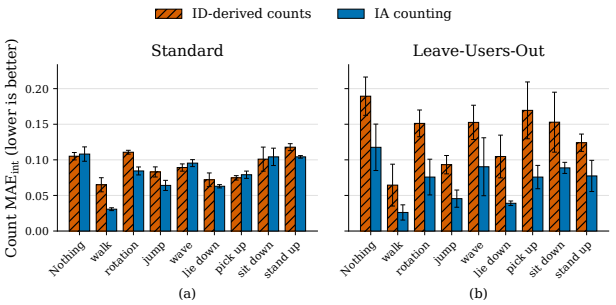


Fig. 4: Per-activity integer-count MAE for both formulations under the (a) Standard and (b) Leave-Users-Out protocols (5 GHz, all environments). Error bars denote SD over three data partitions.

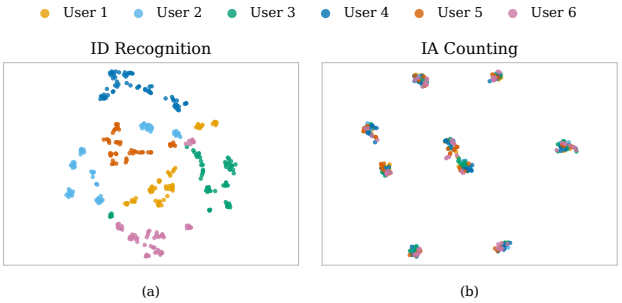


Fig. 5: Separately fitted t-SNE projections of identity-dependent and identity-agnostic embeddings, colored by user identity (5 GHz, all environments, Standard protocol, data partition 1, single-user samples).

31.18%, while remaining well away from the all-zero solution on every metric. Training from scratch increased $\mathrm{MAE}_{\mathrm{int}}$ by a factor of 3.5 to 0.2849 and yielded an $R^2_{\mathrm{int}}$ of $-0.2002$; as in the identity-agnostic implementation of THAT (Section V-A), this configuration assigned zero to every activity count.

## VI. Discussion

### A. Determinants of Pipeline Performance

The ablations reveal three levels of sensitivity. Spatial resolution, interpolation, and channel strategy changed $\mathrm{MAE}_{\mathrm{int}}$ by at most 0.0011. Removing temporal warping or permuting the antenna–subcarrier axis had moderate effects, whereas temporal permutation and training from scratch were most damaging. The pipeline is therefore robust to minor projection choices but relies on temporal locality and pretrained initialization. Temporal permutation still outperformed the all-zero predictor, indicating that useful scene information remained despite the disrupted locality. By contrast, training from scratch and the identity-agnostic implementation of THAT produced identical count metrics and all-zero rounded outputs ($\mathrm{MAE}_{\mathrm{int}} = 0.2849$), suggesting that zero-inflated targets can induce the same degenerate solution across models of different capacities under the evaluated optimization setting.

Combined-environment training outperformed environment-specific training, but this may reflect both its roughly threefold larger training set and exposure to diverse multipath structures. The consistent 5 GHz advantage is expected: the wider channel bandwidth gives finer frequency resolution and the shorter wavelength increases sensitivity to small-scale body displacement.

### B. Why Counting Outperforms Identity-Indexed Prediction

The two formulations share the same preprocessing and backbone and differ only in output head and objective, isolating the effect of problem formulation. Despite receiving coarser supervision, identity-agnostic counting is more accurate on every count metric except exact match, and this ordering holds in each of the eight environment–band conditions. The separation widens sharply under user shift. There the derived counts are barely more accurate than predicting zero for every activity, while identity-agnostic counting attains roughly half that error. This difference is not visible under closed-set evaluation.

The training objectives provide a plausible explanation. User-indexed prediction must determine who performs each activity, encouraging features linked to body-induced multipath signatures and habitual motion patterns. These features can aid enrolled-user recognition but become unreliable under user shift, and prioritising them may divert representational capacity from activity-composition structure. Counting instead assigns the same target to a given activity composition regardless of which users perform the activities. Its loss therefore does not explicitly reward representations that separate users.

The representation analyses support this interpretation. Identity-dependent embeddings cluster primarily by user identity, whereas identity-agnostic embeddings cluster by activity composition and show substantial user mixing. With identical frozen linear counting heads, identity-agnostic features yield 26% lower counting error than identity-dependent features in-distribution and 23% lower under user shift. Because both probes used the same readout and objective, the remaining gap cannot be attributed to the final prediction head; it reflects differences already encoded in the representations by the original training formulations. Environment shift remains challenging because room-dependent multipath changes also affect activity-related components; as fractions of the zero-predictor error the gap is smaller under LOEO (0.66 against 0.88) than under LUO (0.51 against 0.96). The widening gap at five users is consistent with the additional slot-to-user assignment burden of identity-indexed prediction.

### C. Metric Behaviour and Error Sources

The higher closed-set exact-match accuracy of the derived counts does not imply more accurate counting, since their cell-wise accuracy is lower and their MAE is higher. The error distribution reported in Section V-B explains the difference: derived count errors are concentrated in fewer samples but affect more entries within them. Substituting one activity for another changes two count-vector entries, whereas an activity–$\emptyset$ confusion changes one; conversely, permuting activities between user slots penalises identity recognition while leaving the aggregate count vector unchanged. These mechanisms explain how the derived counts can lead in closed-set exact-match accuracy despite worse cell-wise counting performance.

Rounding generally reduces MAE because most continuous predictions lie closer to the correct integer, but it can lower $R^2$ when predictions cross integer boundaries. We therefore

treat $\mathrm{MAE}_{\mathrm{int}}$ as the primary metric because it evaluates both formulations in the same integer count space, is directly interpretable as the mean count error per activity–sample entry, and is not normalized by protocol-specific target variance.

Both formulations struggle with similar activities, indicating a shared component of signal ambiguity without excluding formulation-specific effects. *Sit down* and *stand up* involve similar vertical displacements distinguished mainly by temporal order, while distinguishing *Nothing* from $\emptyset$ depends on static multipath cues separating a stationary user from an absent one. *Nothing*'s confusion with *walk* and *stand up* is consistent with these activities ending in a stationary posture. *Rotation*, *walk*, and *jump* also share localised motion signatures. Error increases with occupancy as concurrent movements produce overlapping channel effects, while the near-zero error for empty scenes indicates that the absence of activity is reliably detected.

### D. Practical Implications and Limitations

The identity-dependent formulation ties output slots to physical individuals, requiring prior enrolment and retraining when the population changes. Identity-agnostic counting instead produces a fixed-length vector determined only by the activity set and can operate without a predefined user roster. Its output dimensionality is independent of the maximum supported occupancy, although reliable count ranges remain bounded by the occupancies observed during training. Summing the predicted activity counts also provides an implicit occupancy estimate without a separate prediction head. For deployment, parameter and MAC reductions did not translate proportionally into CPU latency, so hardware-specific runtime and memory measurements are more informative than complexity proxies alone.

Because its output describes the scene rather than specific individuals, counting avoids explicit identity linkage at the point of reporting: a count vector specifies how many users perform each activity without identifying those users. This output-level data minimisation is not a formal privacy guarantee, since the underlying CSI and learned representations may still contain recoverable user-specific information. Counting therefore complements rather than replaces signal- or representation-level privacy mechanisms. It is well suited to aggregate activity and occupancy monitoring, but cannot by itself provide the event-to-person attribution required in personalised healthcare, assisted living, access control, and individual-level safety monitoring.

The evaluation is confined to WiMANS, which, to our knowledge, remains the only publicly available dataset for multi-user WiFi activity sensing. Its recordings therefore share a single hardware platform and capture configuration, and the flattened representation used here is tied to that configuration's antenna and subcarrier counts. Within these limits the lower-MAE ordering is consistent, repeating across all eight environment–band conditions and the Standard, LUO, and LOEO protocols, but whether it transfers to other deployments remains open. Two boundaries of the setting are worth noting. Scenes contain at most five concurrent users, and the gap between formulations widens at the highest observed occupancy, so denser scenes are the natural next test rather than a safe extrapolation. Only CSI amplitude is used, since raw phase from commodity devices carries hardware-induced offsets that require sanitisation; recovering it may improve the separation of activities with similar amplitude signatures, such as *sit down* and *stand up*.

## VII. Conclusion

This work compared identity-dependent recognition with identity-agnostic activity counting using matched integer-valued count metrics and a shared pipeline that isolates the effect of output structure and objective. Despite using coarser supervision, identity-agnostic counting achieved lower MAE, higher cell-wise accuracy, and higher explained variance under closed-set conditions, repeating across all eight environment–band conditions; derived counts led only in exact-match accuracy, reflecting a different concentration of errors rather than lower overall count error. The two formulations separated further under domain shift, where identity-agnostic counting led on every counting metric under both unseen users and unseen environments, and the derived counts remained close to the all-zero predictor under unseen users. Closed-set evaluation therefore overstates how well identity-indexed prediction recovers scene-level activity composition. Representation analyses indicate why: identity-dependent embeddings were organized primarily by user identity and identity-agnostic embeddings by activity composition, and a linear probe on frozen features localized the difference to the representation rather than the output head. Requiring a model to decide who performs each activity is therefore not a neutral design choice, since it commits capacity to structure that is informative only for enrolled users. Where the question is what is happening in a scene rather than who is doing it, activity counting is the more accurate and robust formulation, and operates without a predefined user roster; extending it to denser scenes and other capture configurations is the natural next step.

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